

The Electron Unit Charge in T0: Geometric Resolution of Point Singularities

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- α with $E_0 = 7.400 \text{ MeV}$: 7.300×10^{-3} , $1/\alpha = 137.00 \rightarrow 7.3013 \times 10^{-3}$, $1/\alpha = 136.96$ (with $E_0 = 7.398 \text{ MeV}$: 137.035).
- Born-Infeld: $\beta \approx 10^{18} \text{ V/m}$, $r_{\text{eff}} \approx 1/\beta$, $U \approx e^2 \beta / (4\pi\epsilon_0) \rightarrow \beta \approx 1.2 \times 10^{20} \text{ V/m}$, $r_0 = \sqrt{e / (4\pi\epsilon_0 \beta)} \approx 3.5 \times 10^{-15} \text{ m}$, $U \approx 1.24 e^2 / (4\pi\epsilon_0 r_0)$.
- Charge quantization $q_n = -n e \xi^{1/2}$: annotated $-n = 1$ gives $-0.01155 e$, not $-e$.

Abstract

The classical representation of the electron unit charge as a point singularity encounters fundamental issues in quantum electrodynamics (QED), such as infinite self-energy and ultraviolet divergences. This treatise, authored as the creator of T0 Theory (Time-Mass Duality Framework), demonstrates how T0 resolves these singularities by treating charge as an emergent, geometric property of a universal field. Based on the single parameter $\xi = \frac{4}{3} \times 10^{-4}$ and the Time-Mass Duality $T_{\text{field}} \cdot E_{\text{field}} = 1$, the charge is derived as a fractal pattern of quantized scales (effective fractal dimension $D_f^{\text{eff}} \approx 2.973$). This avoids infinities, explains observations like the fine-structure constant $\alpha \approx 1/137$, and seamlessly connects to kinematic models in Electromagnetic Mechanics. The GitHub documentation for T0 Theory (current as of October 21, 2025) serves as a reference for detailed derivations.

.1 Introduction: The Problem of Point Singularities

In standard physics, the electron unit charge $-e \approx -1.602 \times 10^{-19}$ C is modeled as a Dirac delta function $\rho(\mathbf{r}) = -e\delta(\mathbf{r})$. This leads to a Coulomb field $E(\mathbf{r}) \propto 1/r^2$ and infinite electrostatic self-energy:

$$U = \frac{1}{2} \int \epsilon_0 E^2 dV \rightarrow \infty \quad (\text{as } r \rightarrow 0). \quad (1)$$

QED addresses this through renormalization (vacuum polarization), yet the bare point singularity remains a mathematical artifact. Experimentally, the electron appears point-like (to $< 10^{-22}$ m), but this does not preclude extended models at deeper scales. T0 Theory, which I developed as its creator, radically resolves this dilemma: Charge is not an intrinsic point property but an emergent projection of geometric patterns in the universal field.

.2 Alternative Representations of Charge

Nonlinear Electrodynamics

In models like Born-Infeld, the field saturates at maximum strength $\beta \approx 1.2 \times 10^{20}$ V/m, yielding an effective charge radius $r_0 = \sqrt{e/(4\pi\epsilon_0\beta)} \approx 3.5 \times 10^{-15}$ m. This results in finite self-energy $U \approx 1.24 e^2/(4\pi\epsilon_0 r_0)$. *(Corrected on 30 Sept. 2026: previously $\beta \approx 10^{18}$ V/m, $r_{\text{eff}} \approx 1/\beta$ and $U \approx e^2\beta/(4\pi\epsilon_0) - 1/\beta$ has the unit m/V and is not a length, $e^2\beta/(4\pi\epsilon_0)$ has the unit J·V and is not an energy. In Born's model r_0 is fixed by $e/(4\pi\epsilon_0 r_0^2) = \beta$; with $U = m_e c^2$ one obtains $r_0 = 3.48 \times 10^{-15}$ m and $\beta = 1.19 \times 10^{20}$ V/m.)*

Soliton and Vortex Models

The electron as a stable wave packet in nonlinear field theories (e.g., sine-Gordon) distributes the charge density $\rho(r)$ over a finite width, with $E \propto q(r)/r^2$ and $q(r) \rightarrow 0$ as $r \rightarrow 0$.

Topological Defects

Charge as a Chern-Simons vortex in gauge theories, quantized by topology ($\pi_3(S^2) = \mathbb{Z}$), without a bare singularity.

Model	Singularity?	Self-Energy
Point Charge (QED)	Yes	∞ (renormalized)
Born-Infeld	Effectively no	Finite
Soliton	No	Finite (from field energy)
T0 Geometry	No	From ξ -scaling

Table 1: Comparison of alternative charge representations

.3 The Electron Charge in T0 Theory

Time-Mass Duality and Emergence

T0 Theory unifies quantum mechanics and relativity in a parameter-free framework via $T_{\text{field}} \cdot E_{\text{field}} = 1$. Particles emerge as excitation patterns in the field, governed by $\xi = \frac{4}{3} \times 10^{-4}$. The fine-structure constant arises as:

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2, \quad E_0 = 7.400 \text{ MeV}, \quad (2)$$

yielding $\alpha \approx 7.3013 \times 10^{-3}$ ($1/\alpha \approx 136.96$)—with fractal corrections for the exact CODATA value 137.035999177. *(Corrected on 30 Sept. 2026: previously 7.300×10^{-3} and $1/\alpha \approx 137.00 - 1.33333 \times 10^{-4} \times 7.400^2 = 1.33333 \times 10^{-4} \times 54.76 = 7.3013 \times 10^{-3}$, i.e. $1/\alpha = 136.96$; with the fractally corrected $E_0 = 7.348/\sqrt{K_{\text{frac}}} = 7.398 \text{ MeV}$ (Doc. 190, R72) one gets 137.035.)*

The charge $-e$ is a dimensionless geometric relation: $q^{\text{T0}} = -1$ (in natural units), projected via $S_{\text{T0}} = 1.782662 \times 10^{-30} \text{ kg}$ onto SI values. No singularity, as the charge density is fractally distributed:

$$\rho(r) \propto \xi \cdot f_{\text{fractal}} \left(\frac{r}{\lambda_{\text{Compton}}} \right), \quad (3)$$

with $f_{\text{fractal}}(r) = \prod_{n=1}^{137} \left(1 + \delta_n \cdot \xi \cdot \left(\frac{4}{3} \right)^{n-1} \right)$ and effective fractal dimension $D_f^{\text{eff}} \approx 2.973$.

Finite Self-Energy and Quantization

The self-energy is finite:

$$U = \frac{1}{2} \int \epsilon_0 E^2 dV = \frac{e^2}{8\pi\epsilon_0 r_e} \cdot K_{\text{frac}}, \quad (4)$$

$$r_e \approx 2.817 \times 10^{-15} \text{ m} \quad (\text{classical radius from } \xi\text{-scaling}), \quad (5)$$

$$K_{\text{frac}} = 0.986 \quad (\text{fractal correction factor}). \quad (6)$$

Quantization follows from discrete scales: $q_n = -n \cdot e \cdot \xi^{1/2}$, with $n = 1$ for the unit charge. *(Note of 30 Sept. 2026: With $\xi^{1/2} = \sqrt{4/30000} = 0.011547$, $n = 1$ gives the charge $q_1 = -0.01155 e$, not $-e$; the formula yields the unit charge only for $n = \xi^{-1/2} = 86.6$, i.e. for no integer n .)* This aligns with topological quantization (Chern number = 1), ensuring stability without collapse.

.4 Implications for Electromagnetic Mechanics

T0 integrates with kinematic mechanics: Charge emerges as a rotating EM vortex, stabilized by fractal correction. No Dirac delta— $\rho(r)$ is a helical pattern, enabling singularity-free simulations. Applications: g-2 anomaly predictions and LHC mass spectra.

.5 Notation

ξ Geometric parameter; $\xi = \frac{4}{3} \times 10^{-4}$

S_{T0} Scaling factor; $S_{\text{T0}} = 1.782662 \times 10^{-30} \text{ kg}$

f_{fractal} Fractal function; $\prod_{n=1}^{137} (1 + \delta_n \cdot \xi \cdot (4/3)^{n-1})$

D_f^{eff} Effective fractal dimension; $D_f^{\text{eff}} \approx 2.973$ (local: $D_f^{\text{space}} = 3 - \xi \approx 2.9999$)